Lecture 8: The Benefits of Competition & How to Produce (Chapters 7 · 3.5)

Applied Microeconomics — Master Notes · by Jakob V. Stangel

Required reading: Friberg, **Ch. 7 — The benefits of competition** (consumer surplus, producer surplus, social welfare, deadweight loss, policy) · **§3.5 — How to produce** (cost minimisation: isoquants, isocost, MRTS, the Lagrangian). *Lecturer: L. Butera — decks "**The benefits of competition**" + "**What inputs to use to produce?**" (Two readings combined. **Ch. 7 is the new, central part** — it finally says why we care that markets are competitive. **§3.5 is the long-run cost-minimisation machinery**, most of which you already have from Lecture 6; here it's recapped + the calculus added.)*

Résumé. Lecture 7 showed competition drives price to **min AC** and profit to zero. This lecture asks the "**so what?**" — *who gains, and by how much?* The answer is in **surplus**: the demand curve is **willingness to pay**, so the gap between what consumers *would* pay and what they *do* pay is **consumer surplus**; the gap between the price sellers *get* and the minimum they'd *accept* is **producer surplus**. Their sum is **social welfare (W = CS + PS)**, and the competitive equilibrium **maximises** it — any distortion (a tax, a quota, a limit on firms) opens a **deadweight loss**. The second reading, **§3.5**, is the *cost-minimisation* side of the firm already met in Lecture 6: choose (L*, K*) where the **isoquant is tangent to the lowest isocost** — now stated as three equivalent rules and solved with a **Lagrangian**.

> ★ **The heart of the lecture — competition maximises total surplus.**
> • **Consumer surplus (CS)** = area **under the demand curve, above the price** (what you'd have paid − what you did pay).
> • **Producer surplus (PS)** = area **above the supply curve, below the price** (what you got − the minimum you'd accept). **PS = Revenue − Variable cost**; it equals **profit** whenever fixed costs are zero (the long run).
> • **Social welfare W = CS + PS** is **largest at the competitive equilibrium.** Shove the market away from it and the lost surplus that *no one* captures is the **deadweight loss**.
> • **How to produce (§3.5):** for a target output, minimise cost at the **tangency of isoquant & isocost** — the **last-dollar rule** $MP_L/w = MP_K/r$ (same optimum as the consumer's, mirror-imaged).

Red thread: *(1) How much does competition actually benefit consumers and producers — and how do we measure it? (2) What does a government policy do to that total benefit? (3) [§3.5] Given a target output, what's the cheapest input mix?*

Key concepts / "modes" to use: *consumer surplus (area under demand, above price) · marginal willingness to pay · producer surplus (area above supply, below price); **PS = R − VC**; PS = profit when F = 0 · social welfare **W = CS + PS** · deadweight loss (DWL) · surplus & elasticity (inelastic ⇒ bigger surplus at stake) · the two ways policy bites (shift a curve · drive a wedge p ≠ MC) · cost minimisation: isoquant · isocost (wL + rK = C) · MRTS · the lowest-isocost / tangency / last-dollar rules · the Lagrangian.*

Why competition benefits society (the set-up)

Core: we proved in Lecture 7 that long-run competition pins price to **min AC** and profit to **zero**; this lecture measures the **gain that creates for everyone else.**

- *Recap of the mechanism:* a short-run profit (say from a **scarce input** — a prime location, a star manager) gets **bid away** as rivals compete for that input; with identical firms the long-run **supply curve is flat**, and firms must minimise cost just to survive.
- The result is cheap, abundant output — and the tool for measuring "how good is that for society?" is **surplus.** *(Welfare here means **efficiency** — the size of the total pie — not fairness/distribution.)*

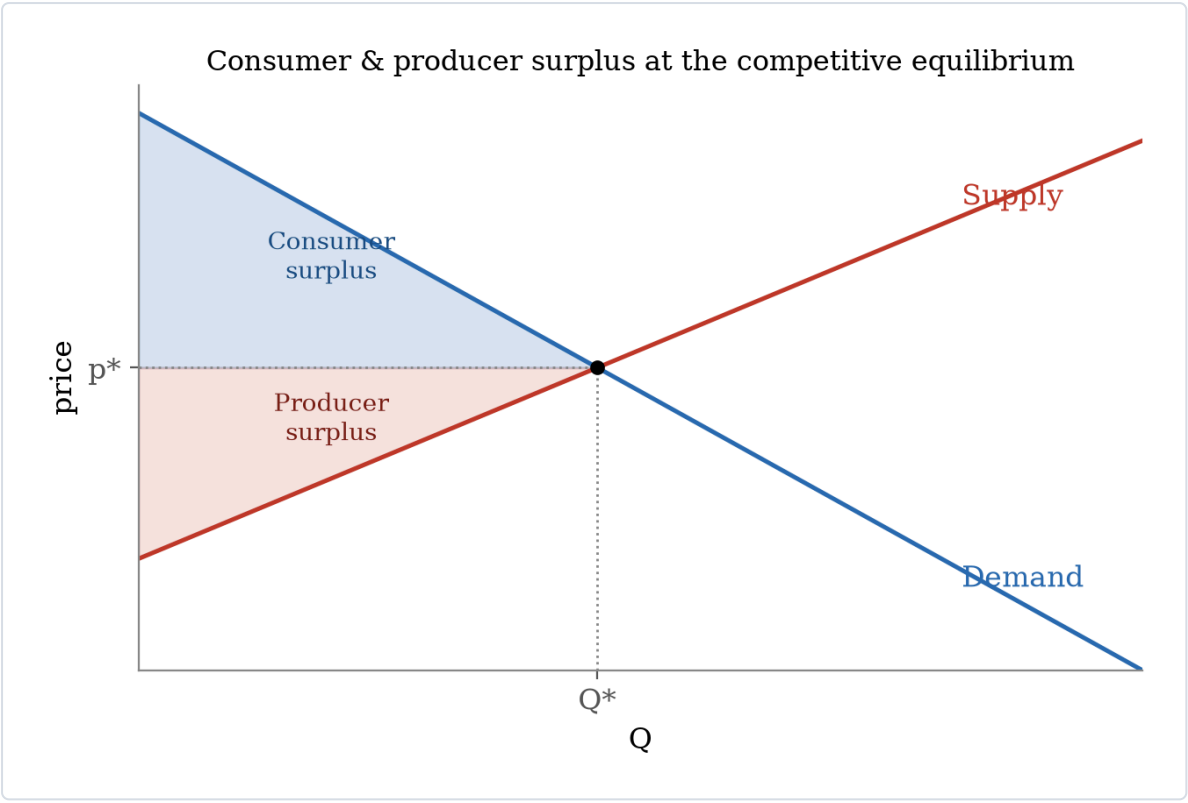
★ Consumer surplus (CS)

Core: the **demand curve is a willingness-to-pay curve**; consumer surplus is the **bargain** consumers get — what they *would* have paid minus what they *actually* pay.

- **The demand curve = marginal willingness to pay:** its height at each unit is the **most** a consumer would spend for that extra unit — the **marginal value** they place on it.
- **Consumer surplus** = (willingness to pay for a quantity) − (what they actually pay for it), summed over all units bought.

> CONSUMER SURPLUS
>
> $\mathrm{CS} = \text{area}$ **under the demand curve** and **above the market price** $(\text{up to } \mathrm{Q})$

Reads: **CS** is the triangle-ish region between the **demand curve** (what each unit is worth to buyers) and the **horizontal price line** (what they pay), out to the quantity bought. · Every unit to the left of Q* was worth **more** than its price to someone — that stacked-up "extra value" is the surplus.



How to read it.

- **Axes:** price (y) vs quantity (x); **Demand** slopes down, **Supply** slopes up, crossing at the **competitive equilibrium (Q*, p*)**.
- **Consumer surplus** = the **upper** shaded wedge (under demand, above p*) — buyers' total "bargain."
- **Producer surplus** = the **lower** shaded wedge (above supply, below p*) — sellers' total "bonus" over their minimum acceptable price.
- Together they fill the whole area between the curves up to Q* — that's **total surplus / welfare**.

★ Producer surplus (PS)

Core: the mirror image — the gap between the price sellers **receive** and the **minimum** they'd have accepted (which the supply curve traces, since supply = marginal cost).

- **Producer surplus** = (what the good sells for) − (the minimum needed to make the seller willing to produce it), i.e. the area **above the supply curve and below the price**, up to Q.
- Because the supply curve is **marginal cost**, that area equals **revenue minus variable cost**:

PRODUCER SURPLUS

$$\mathrm{PS} = \mathrm{R} - \mathrm{VC}$$

Reads: **PS** = total **revenue** minus **variable cost** — the area above the (marginal-cost) supply curve, below price. · it is *not* the same as profit…

- **PS vs profit:** profit = R − VC − **FC**. So **PS − profit = FC** (fixed cost). Therefore:

SURPLUS VS PROFIT

$$\text{PS} = \text{profit} + \text{FC} \Rightarrow \text{if FC} = 0 \text{ (long run)}, \textbf{PS = profit}$$

Reads: producer surplus exceeds profit by exactly the **fixed cost F**; when there are no fixed costs (the long run), **producer surplus and profit coincide.** · (So in the long run, "zero profit" ≠ "zero producer surplus" only if there were fixed costs to recover — with F = 0 they're identical.)

★ Social welfare = CS + PS

Core: add the two surpluses and you get society's total gain from the market — and the **competitive equilibrium makes it as big as possible.**

SOCIAL WELFARE

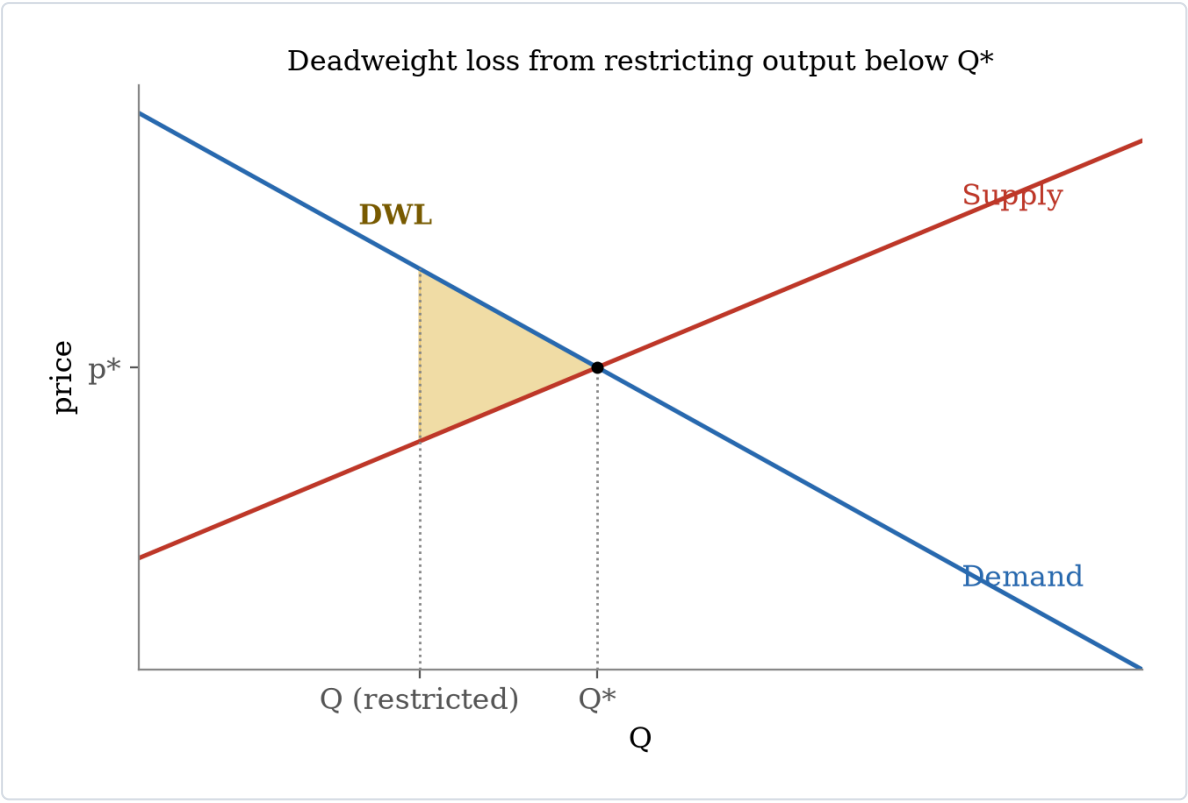
$$\text{W} = \text{CS} + \text{PS}$$

Reads: total welfare **W** = consumer surplus **plus** producer surplus — the **whole** area between the demand and supply curves up to the quantity traded. · at the competitive **(Q*, p*)** this area is **maximised**: every unit whose value (demand) exceeds its cost (supply) gets produced, and no unit whose cost exceeds its value does.

★ Deadweight loss (DWL)

Core: any move away from the competitive equilibrium **destroys** some surplus that **no one** picks up — that lost welfare is the deadweight loss.

Deadweight loss = the **net reduction in total welfare** when a policy or distortion shifts the market off (Q*, p*): one group loses surplus, and that loss is **not** fully offset by anyone else's gain.



How to read it.

- Output is forced **below Q*** (e.g. a quota or a cap on firms). On every unit between the restricted quantity and **Q***, buyers valued the good (demand) **more** than it cost to make (supply).
- Those mutually-beneficial trades **don't happen** → the shaded **DWL triangle** of surplus simply **vanishes** (it's not transferred to anyone — it's pure loss).
- The triangle's **width** = how far output is pushed from Q*; the bigger the distortion, the bigger the DWL.

Surplus & elasticity

Core: surplus is tied to **elasticity** — the **more inelastic** your demand for a good, the **more** you benefit from it being sold in a competitive market (and the more you lose if its price rises).

- Intuition: if you *must* have it (**inelastic**), the gap between your high willingness-to-pay and the low competitive price is **huge** → big consumer surplus. If you can easily do without it (**elastic**), that gap is small.
- **The deck's comparison** — a price rise from p1 to p2, elastic vs inelastic demand:

	Relatively elastic D	**Relatively inelastic D**
CS at **p1** (low price)	A + C	A + B + C + D
CS at **p2** (high price)	A	A + B
CS lost from the price rise	**– C**	**– C – D**

→ The **inelastic** consumer loses **more** (C + D vs just C) when price rises — they had more surplus at stake. *(Same logic sits behind tax incidence in Lecture 3: the inelastic side bears more.)*

Policy applications

Core: essentially **every** government action hits a competitive market in one of **two** ways.

How policy bites	What it does	Examples
1. Shifts a curve	moves supply or demand, changing (Q*, p*)	**limits on the number of firms** (e.g. rules slowing how fast firms can exit, to protect workers); **quotas / output limits**
2. Drives a wedge	makes **price ≠ marginal cost** even though they were equal in the competitive equilibrium	a **tax** (the tax wedge from Lecture 3), a subsidy, a price control

→ Either way, if it pushes the market off (Q*, p*), it generally creates a **deadweight loss** — the analytical point of the whole chapter is to be able to **draw the policy, shade the lost surplus, and name who gained and who lost.**

How to produce: cost minimisation (§3.5)

Core: the long-run "**how to produce**" problem — pick the input mix (L*, K*) that makes a target output $\bar{q}$ as **cheaply** as possible. *(You met this in Lecture 6; here is the compact statement + the calculus.)*

- **The problem = the consumer's, mirror-imaged.** Objective = the **production function Q(L, K)** (like utility U); constraint = the **isocost C = wL + rK** (like the budget), where **w** = wage, **r** = rental rate of capital.
- **Isoquant** = all (L, K) giving the **same output** (the firm's indifference curve). **Isocost** = all (L, K) costing the **same** total (the firm's budget line); its slope is **−w/r**.
- **MRTS** (marginal rate of technical substitution) = slope of the isoquant = how much K one extra L replaces at constant output = **MP_L / MP_K**.
- **Three equivalent cost-minimisation rules:**
  1. **Lowest-isocost rule** — pick where the **lowest** isocost line just **touches** the target isoquant.
  2. **Tangency rule** — pick where the isoquant is **tangent** to the isocost: **MRTS = w/r**.
  3. **Last-dollar rule** — the last dollar on each input yields the **same** extra output:

COST-MINIMISING INPUT MIX

$$\text{MRTS} = \text{MP}_\text{L}/\text{MP}_\text{K} = \text{w}/\text{r} \Leftrightarrow \text{MP}_\text{L}/\text{w} = \text{MP}_\text{K}/\text{r}$$

Reads: at the optimum the **marginal product per dollar** is equal across inputs — spend the last dollar wherever it buys the most output, and in equilibrium that's the same everywhere. · **w** = wage (price of labour) · **r** = rental rate (price of capital).

- **The calculus (Lagrangian).** Minimise cost subject to hitting output $\bar{q}$:

COST-MINIMISATION LAGRANGIAN

$$\mathcal{L} = wL + rK + \lambda \cdot (\bar{q} - Q(L, K))$$

Reads: minimise the **cost** wL + rK subject to the **output constraint** Q(L,K) = $\bar{q}$; the multiplier **λ** comes out as **marginal cost** (the extra cost of one more unit of output). The first-order conditions reproduce the **last-dollar rule** above. *(Exactly analogous to the consumer's Lagrangian in Lecture 5 — and the **dual** of maximising output for a given budget.)*

Formula sheet (Lecture 8)

Concept	Formula	In words
Consumer surplus	CS = area under demand, above p	buyers' total "bargain"
Producer surplus	**PS = R – VC**	sellers' bonus above min acceptable price
PS vs profit	PS = profit + FC (⇒ PS = profit if F = 0)	they coincide in the long run
Social welfare	**W = CS + PS**	total surplus; max at competitive eq.
Deadweight loss	lost (CS + PS) from a distortion	surplus no one captures
Cost-min (tangency)	MRTS = MP_L/MP_K = **w/r**	isoquant tangent to isocost
Cost-min (last-dollar)	**MP_L/w = MP_K/r**	equal marginal product per dollar
Cost-min (Lagrangian)	$\mathcal{L}$ = wL + rK + λ($\bar{q}$ – Q(L,K))	λ = marginal cost

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★ Fun facts & memorable details (Lecture 8: Benefits of Competition)

> The sticky bits from the welfare lecture.

- **The demand curve is a willingness-to-pay curve in disguise.** Its height over each unit is the **most** someone would pay for it — so the area between it and the price is literally the stack of **bargains** buyers walk away with.
- **Producer surplus ≠ profit (until the long run).** They differ by exactly the **fixed cost**: PS = profit + F. Sell off the fixed assets (long run, F = 0) and the two become the **same number**.

- **Deadweight loss is surplus that evaporates.** Unlike a tax transfer (surplus moves from you to the government), DWL goes to **no one** — it's the value of mutually-beneficial trades that simply never happen.
- **The needier you are, the more competition does for you.** Inelastic demand = a big gap between your willingness to pay and the competitive price = **large consumer surplus**. Competition quietly helps most on the things you can't do without.
- **Almost every policy is "shift a curve" or "drive a wedge."** Taxes, quotas, price controls, licensing — reduce the whole zoo to those two moves, draw it, and shade the deadweight-loss triangle. That's the exam skill.
